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Dear Editor,

We are pleased to submit our manuscript entitled “**SkelGym: Cross-Paradigm Kinematic Fusion and Dynamic Augmentation for 22-Class Gym Exercise Classification**” for consideration for publication in *Expert Systems with Applications*.

Why Expert Systems with Applications

SkelGym represents an end-to-end **intelligent exercise recognition system** with direct practical deployment value, making it well suited to the scope of ESWA. Rather than presenting a purely theoretical deep learning study, our work delivers a complete automated pipeline—from raw monocular video input through 3D skeletal extraction, biomechanical feature engineering, physiologically valid data augmentation, multi-paradigm classification, to optimized ensemble fusion and real-time edge deployment—that addresses the practical challenge of automated virtual personal training in home and commercial gym environments.

Novelty and Contributions

The manuscript introduces three core innovations:

1. **117-dimensional Biomechanical Compound Representation:** A compact kinematic descriptor unifying 3D relative joint displacements with pairwise elevation and azimuth angles, resolving the multicollinearity of combinatorial 3-joint angle triplets while remaining computationally lightweight.
2. **SkelGym-Aug:** An empirically selected, task-oriented skeletal data augmentation configuration specifically tailored for resistance exercise biomechanics, determined via systematic Leave-One-Out ablation to enforce geometric invariance while preserving exercise cadence.
3. **Validation-Calibrated Cross-Paradigm Ensemble:** A principled SLSQP-optimized soft voting formulation coupling self-attention Transformers with four-stream Adaptive Graph Convolutional Networks, with calibration performed strictly on validation partitions to prevent test-set contamination.

Practical Application Value

SkelGym achieves $69.74\% \pm 1.04\%$ window-level accuracy and $79.11\% \pm 0.25\%$ video-level consensus accuracy (Video Macro F1: 0.7834 ± 0.0082 , Window Macro F1: 0.6882 ± 0.0068) across three independent random seeds across 22 fine-grained resistance exercises with a single-window classifier

inference latency of merely 0.42–4.33 ms on CPU (0.08–0.54 ms on CUDA, 1.11–8.77 ms on MPS; where single-model backbones execute in 0.42–1.16 ms and the full five-stream ensemble executes in 4.33 ms). By operating exclusively on privacy-aware 3D skeletal coordinates, the system eliminates the need for continuous RGB video streaming, making it suitable for privacy-sensitive deployment on smart mirrors, mobile phones, and embedded gym displays. Selected model comparisons were evaluated via paired statistical tests (McNemar $p < 10^{-12}$, Wilcoxon $p < 0.005$ against $\alpha_{\text{adj}} = 0.01$), and a comprehensive biomechanical error taxonomy identifies the physical limitations of monocular pose estimation.

Originality Statement

This manuscript has not been previously published and is not under simultaneous consideration for publication elsewhere. All authors have read and approved the final manuscript. The work is original and all sources are properly cited. All source code, extracted skeletal datasets, metadata, and pre-trained model checkpoints are publicly available at <https://huggingface.co/Cuong2004/gym-exercise-classification>.

We believe that the combination of practical deployment feasibility, rigorous statistical validation, and open-science reproducibility makes this work a strong fit for the readership of *Expert Systems with Applications*. We look forward to your consideration.

Sincerely,

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